

# Medicare Hospital Value Scorecard

One-page memory notes + execution plan (CMS Hospital data)

## 1. Project in one blink

**Question:** Which hospitals deliver better patient experience + clinical quality for the Medicare dollars spent — and where should a payer / health system focus improvement?

**Answer shape:** Build one hospital-level table (Facility ID) with Experience, Readmission proxy, Infection safety, MSPB cost, then a Value Score = Quality / Spending. Benchmark by state. Flag high-quality/high-cost and low-quality/high-cost outliers. Dashboard filters: State, Ownership, Size proxy.

## 2. Do you have the right CSVs?

**YES — you already have all REQUIRED files.** Nothing else is mandatory to finish P1–P6.

| File you have                                       | Role in project                                   | KPI it feeds              |
|-----------------------------------------------------|---------------------------------------------------|---------------------------|
| HCAHPS-Hospital.csv                                 | Patient experience survey (star + linear scores)  | Experience score          |
| Unplanned_Hospital_Visits-Hospital.csv              | 30-day readmissions + Hybrid HWR                  | Readmission proxy         |
| Healthcare_Associated_Infections-Hospital.csv       | Infection SIRs (CLABSI, CAUTI, SSI, MRSA, C.diff) | Safety / quality piece    |
| Medicare_Hospital_Spending_Per_Patient-Hospital.csv | MSPB-1 ratio vs national                          | Spending (cost)           |
| Hospital_General_Information.csv                    | Name, state, ownership, type, overall rating      | Filters + geography grain |

**Missing REQUIRED downloads:** None.

**Optional (not required):** Complications & Deaths (mortality), Payment — Hospital (beyond MSPB), Timely & Effective Care, CMS Hospital Data Dictionary PDF. Skip unless you want extra quality dimensions.

**Size filter note:** CMS General Info has no bed count. Use **Hospital Type** (Acute Care vs Critical Access) or HCAHPS survey volume as size proxy. No extra file needed.

## 3. How the pieces connect

All files join on **Facility ID** (CMS CCN / provider ID). Grain for analysis = **one row per hospital** (after pivoting measure-level rows). Geography grain = **State** (primary); County optional.

Raw shape: HCAHPS / HAI / Unplanned are long (many measure rows per hospital). MSPB is ~1 row/hospital. General Info is 1 row/hospital. Pivot → join → analyze.

## 4. KPI definitions (lock these in P1)

| KPI               | Source field / measure                                                                                                    | Direction     |
|-------------------|---------------------------------------------------------------------------------------------------------------------------|---------------|
| Experience        | H_STAR_RATING (overall) or avg of domain STAR / LINEAR scores (H_COMP_*, H_HSP_RATING_LINEAR_SCORE, H_RECND_LINEAR_SCORE) | Higher better |
| Readmission proxy | Prefer Hybrid_HWR Score; else avg of READM_30_AMI / HF / PN / COPD / HIP_KNEE / CABG                                      | Lower better  |
| Infection safety  | Avg of HAI_*_SIR (HAI_1..6). SIR < 1 better than expected                                                                 | Lower better  |
| MSPB ratio        | MSPB-1 Score. 1.00 = national avg. 1.05 = 5% more spend                                                                   | Lower better  |
| Value Score       | Quality composite ÷ MSPB (see formula below)                                                                              | Higher better |

## 5. Value Score formula (P5 — document this)

Step A — flip “lower is better” metrics so all run 0–100 (higher = better):

Exp\_n = percentile\_rank(HCAHPS star or linear) × 100

Readm\_n = (1 – percentile\_rank(readmission rate)) × 100

HAI\_n = (1 – percentile\_rank(avg SIR)) × 100

Quality = 0.40×Exp\_n + 0.35×Readm\_n + 0.25×HAI\_n

$\text{Value Score} = \text{Quality} / \text{MSPB\_Score}$

Weights are a project choice — keep them fixed and state them on the dashboard. Hospitals missing MSPB or all quality pieces → exclude from Value ranking (still show in detail table as incomplete).

## 6. Cleaning rules that will save you (P3)

- Treat as missing: Not Available, Not Applicable, N/A, blank, footnotes-only rows.
- Facility ID: zero-pad to 6 characters (MSPB file currently has blank header + unpadded IDs like 10001 → 010001). Rename blank first column to Facility ID before join.
- Join type: LEFT from General Info; inner-join for Value Score cohort (must have Quality + MSPB).
- Keep Start Date / End Date from each file in your data dictionary as refresh/period note (P2).
- Your periods differ by file (e.g. HCAHPS/HAI ~ Jul 2024–Jun 2025; MSPB calendar 2024; Unplanned multi-year). That is normal CMS — document it; do not invent alignment.

# Medicare Hospital Value Scorecard — Execution Plan

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## 7. Phase checklist (do in order)

- [P1 Scope]** Lock KPIs above. Grain = hospital. Geography = State. Filters = State, Ownership, Hospital Type (size proxy). Decide Value weights.
- [P2 Ingest]** Load 5 CSVs. Build data dictionary (column, meaning, type, example). Note refresh/period dates from Start/End Date. Fix MSPB Facility ID header.
- [P3 Clean]** Standardize Facility ID. Convert scores to numeric. Drop/flag NAs. Pivot long → wide (one row/hospital). Join all on Facility ID.
- [P4 Analyze]** State benchmarks (mean/median). Scatter: Quality vs MSPB. Flag outliers (high cost + low quality = focus; high quality + low cost = exemplars). Ownership / type cuts.
- [P5 Model-lite]** Compute Exp\_n, Readm\_n, HAI\_n, Quality, Value Score. Rank hospitals nationally and within state. Document formula on dashboard.
- [P6 Dashboard]** KPI cards (avg Value, avg MSPB, avg Experience, # hospitals). Filters: State, Ownership, Size proxy. Map by state or points. Detail table with export.

## 8. Analysis story you should be able to tell

1. National: distribution of Value Score; how many hospitals above/below 1.0 MSPB.
2. State: which states run high quality at low relative spend.
3. Scatter quadrants: Star (high Q / low \$), Overpriced (high Q / high \$), Focus (low Q / high \$), Underperform (low Q / low \$).
4. Focus list: bottom-decile Value within a chosen state + ownership segment — improvement targets for a payer/health system.

## 9. Suggested final hospital table columns

Facility ID | Facility Name | State | Ownership | Hospital Type | Overall Rating | Experience | Readmission | Avg HAI SIR | MSPB | Quality | Value Score | Value Rank (State) | Value Rank (US)

## 10. Tooling (pick one stack and stick to it)

- Python (pandas) or Power BI / Tableau. Excel only for tiny prototypes — files are large (HCAHPS ~300k+ rows).
- Map: state choropleth of median Value, or hospital points if you add lat/long later (optional geocode — not required).

## 11. Common traps

- Comparing raw readmission % across conditions without averaging carefully — use Hybrid\_HWR when present.
- Averaging HAI SIRs that include confidence-limit rows — use only \*\_SIR measure IDs.
- Using HCAHPS answer percents (Always/Sometimes) instead of STAR/LINEAR — prefer STAR/LINEAR for composites.
- Forgetting MSPB ID padding → failed joins / duplicate “ghost” hospitals.
- Claiming causal “this hospital is bad” — CMS measures are risk-adjusted but still observational; phrase as relative performance.

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Source hub: <https://data.cms.gov/provider-data/topics/hospitals>

Folder: Medicare Hospital Value Scorecard — 5 CSVs already sufficient for full project delivery.

Generated as personal project notes — no watermark.